Tokens on a Timeline: Tangible Correction of Automatic Speaker Diarization for Low-Resource Speech Annotation

Arnab Satyam Roy
Department of Computer Science
American International University–Bangladesh (AIUB)
Dhaka, Bangladesh
arnabsroy9@gmail.com

Ahsanul Haque
Department of Computer Science
American International University–Bangladesh (AIUB)
Dhaka, Bangladesh
ahsanulhaque1231@outlook.com

Abdus Salam
Department of Computer Science
American International University–Bangladesh (AIUB)
Dhaka, Bangladesh
abdus.salam@aiub.edu

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1 Introduction

Long-form speech understanding, comprising both transcription and speaker attribution, underpins applications including media indexing, meeting summarization, accessibility tooling, and legal and forensic audio analysis. Current systems remain optimized for high-resource languages, and their performance on conversational, multi-speaker, low-resource audio is markedly weaker. Recent work on Bangla long-form speech indicates that further gains depend on more annotated diarization data: the diarization stage of an end-to-end pipeline is limited by a small number of annotated recordings (Bhuiyan et al., 2026), and out-of-the-box diarization on Bangla conversational content reaches a diarization error rate (DER) of around 40% on the only public benchmark (Tabib et al., 2026).

This constraint points to an opportunity located in the interaction layer rather than the model. Faster and more reliable human correction tools yield more corrected annotation, which in turn supports better models for under-resourced languages. The contribution proposed here lies in that interaction layer. The diarization model is frozen and treated as a fixed backend, and no modeling contribution is claimed. Bangla provides the motivation and the deployment setting, namely annotation scarcity, while the interaction question concerns the general problem of correcting the temporal output of a learned recognizer.

We describe a tangible user interface in which a physical token represents each speaker and is placed on a printed timeline to correct the output of a frozen pre-trained diarization model. The contribution is an interaction technique, the empirical hypothesis that justifies it, and the study design that will test the hypothesis. Section 2 reviews related work in four threads: the automatic diarization pipeline and the human-facing correction bottleneck, tangible interaction for time-based information, empirical comparisons of tangible and graphical modalities, and the low-resource Bangla speech setting. Section 3 states the gap addressed at the intersection of these threads. Section 4 lists the objectives and hypotheses. Section 5 specifies the proposed system and study design in full.

2 Literature Review

2.1 Automatic speaker diarization and the correction bottleneck

Speaker diarization is the task of partitioning an audio stream into homogeneous segments according to speaker identity (Bredin et al., 2020). Contemporary systems follow a segmentation–embedding–clustering pipeline in which a sliding-window segmenter produces frame-level speaker activities, a per-window embedding model summarises each window into a fixed-dimensional vector, and an agglomerative or spectral clustering groups windows into speaker tracks. The pyannote.audio toolkit is a widely used open-source reference implementation, with pre-trained models for voice activity detection, speaker change detection, overlapped speech detection, and speaker embedding (Bredin et al., 2020). Recent work on the loss formulation shows that switching from the standard multi-label encoding to a powerset multi-class encoding, in which each overlap pattern is a separate class, removes the fragile detection threshold and yields consistent gains on a wide range of benchmarks (Plaquet and Bredin, 2023). Long-form recognition is increasingly delivered by weakly supervised encoder–decoder models such as Whisper (Radford et al., 2023), which has become a common backbone for Bangla long-form ASR (Bhuiyan et al., 2026). The end-to-end Bangla pipelines that combine Whisper and pyannote report substantial improvements over the unadapted baselines, but the same line of work points to annotated data, not modelling, as the limiting factor for further gains.

Recent work has explored four routes to making the correction step cheaper or faster. The first is post-hoc, model-driven correction of the diarization output. He et al. (2026) describe an LLM-assisted workflow in which a user provides brief corrective feedback in natural language, an LLM interprets which span to re-assign, and online speaker enrollments are added to prevent future attribution errors. A complementary direction uses a fine-tuned LLM as a speaker error corrector at the transcript level, taking the initial diarization hypothesis and the ASR transcript as input and producing a corrected label sequence (Kumar et al., 2025; Efstathiadis et al., 2024). The latter line demonstrates that LLM-based correction can markedly improve diarization accuracy on a held-out benchmark, but the gains are constrained to transcripts produced by the same ASR tool that generated the fine-tuning data. Related work reformulates the correction as an end-to-end neural post-processing step that exploits the interaction between the initial speaker activity and the acoustic features (Prokopalo et al., 2022). All of these systems automate the correction step and so remove the human from the loop, but they require supervised training data that is itself scarce for low-resource languages, and they do not address the underlying question of how a human annotator can correct a model output most effectively.

The second direction is human-in-the-loop active correction. In the diarization setting, this is exemplified by incremental collection-level systems in which a human expert answers simple yes/no questions about whether two segments were spoken by the same speaker, and the system updates the clustering hypothesis accordingly (Prokopalo et al., 2022). The questions are selected by an algorithm that aims to minimise human effort while maximising improvement in the diarization error rate, and a stopping criterion caps the total interaction cost. The same line of work generalises the framework to cross-show diarization through a shared speaker database.

The third direction is the development of annotation tools that instrument the cost of correction rather than only the resulting accuracy. Broux et al. (2018) introduced the Human–Computer Interaction Quantity (HCIQ), a weighted sum of correction-action counts that captures the labour of a human correction session without relying on the diarization error rate alone, and they validated it on the REPERE French broadcast-news corpus. The ALLIES corpus, a meta-corpus of French TV and radio recordings that aggregates several earlier evaluation campaigns, was explicitly designed for human-assisted diarization and uses a similar cost-aware protocol (Tahon et al., 2024). More recently, Yamaguchi (2026) presented voxmap-studio, a React-based open-source diarization annotation tool that records edit-operation counts and active time as a first-class output, and used a three-condition preliminary study on the AMI corpus to compare forms of within-modality assistance. A separate semi-automatic toolkit, AnnoTheia, addresses audio-visual speech annotation in low-resource languages by combining active-speaker detection, transcription, and human-in-the-loop verification (Acosta-Triana et al., 2024). The present article is closest in spirit to this line of work: it treats annotation cost and accuracy as joint outcomes, but it varies the interaction modality rather than the form of within-modality assistance.

The fourth direction is the model-level refinement that sits behind any human-facing tool. This includes both loss-function improvements (Plaquet and Bredin, 2023) and end-to-end Bangla pipelines (Bhuiyan et al., 2026). The present article treats the backend as fixed and isolates the interaction question.

2.2 Tangible interaction for temporal media

A separate thread of work embodies time physically. The reacTable (Jordà et al., 2007) demonstrated that tabletop tangible surfaces can host rich temporal interaction through object manipulation. ChronoTape (Bennett et al., 2012) presents a tangible timeline for personal and family history, in which a paper tape is annotated through a reader device. Tquencer (Kaltenbrunner and Vetter, 2018) provides tangible control for time-based media sequencing through passive tokens and overlays. Mahieux et al. (2022) use the hourglass as a tangible metaphor for navigating through space and time in cultural heritage applications, with a co-design study that demonstrates acceptability among museum professionals. These systems author or arrange temporal media; none addresses the correction of output produced by a learned temporal recognizer.

Theoretical grounding for token-based temporal interfaces comes from the token+constraint framework, in which physical tokens represent digital information and physical constraints are mapped to digital operations (Ullmer et al., 2005). The related Tangible Query Interfaces apply the same idea to database query construction (Ullmer et al., 2003). Both build on Ishii and Ullmer's broader "Tangible Bits" vision of bridging bits and atoms through graspable physical artefacts (Ishii and Ullmer, 1997). A complementary thread investigates the design of passive tokens for multi-touch surfaces: Morales González et al. (2016) show that the geometry of a token determines the grasp users adopt, and that the resulting touch pattern on the surface can be recognised with high accuracy, providing a low-cost way to combine tangible and gestural input without instrumenting the surface itself. The present study inherits the design principle that the token itself, not the visual channel, carries the identity anchor, and applies it to a temporal-correction task rather than to multi-touch gesture disambiguation.

None of the systems reviewed above treats the token as a per-instance identity anchor for the purpose of correcting a recogniser's temporal output, which is the operation the present article investigates. The closest relative is the use of physical objects as identity anchors in tangible database queries, but the queries there are constructed from scratch rather than corrected against an automatically generated hypothesis, and the temporal dimension is not central.

2.3 Empirical comparisons of tangible and graphical interaction

A small but methodologically relevant body of work directly compares tangible and graphical user interfaces on a common task. The earliest formal comparison in the HCI literature is Patten and Ishii (2000), who contrasted a token-based TUI with a graphical icon-based GUI on a location-recall task and reported an advantage for the tangible condition. The TUI subjects performed better at recall and were also more likely to use absolute spatial reference frames to organise the blocks, a strategy that correlated with better recall. The work is a methodological model for the present study: a within-subjects comparison on a single task, with both conditions matched in capability and presentation.

A second strand measures the cognitive cost of tangible interaction. Chandrasekera and Yoon (2015) compared a desktop augmented-reality TUI with a desktop virtual-reality GUI on a creative design task and found significantly lower self-reported cognitive load in the TUI condition ($p = 0.045$), measured with the NASA Task Load Index (NASA-TLX). The interpretation offered is that the tangible interface supports epistemic action, the use of physical manipulation to off-load cognitive work, more effectively than the graphical interface. The present article draws on this methodological lineage: it uses NASA-TLX as a secondary outcome and tests a comparable primary question (correction accuracy) on a different task (diarization correction).

A third strand compares tangible and graphical interaction in the context of the token+constraint framework (Ullmer et al., 2003, 2005). Ullmer et al.'s preliminary user study of a tangible database query interface against a comparable graphical interface reported a measurable advantage for the tangible condition on the specific sub-task where physical rotation mapped to parameter value changes; the advantage disappeared for sub-tasks that did not exploit a physical affordance unique to the tangible condition. The present study follows the same logic of designing the comparison around the affordance unique to each modality, and the double-dissociation prediction in Section 4 is an explicit generalisation of the affordance-specific-advantage pattern.

2.4 Low-resource Bangla speech as the deployment setting

The deployment context for the proposed work is Bangla, a low-resource language in the long-form speech regime. Public Bangla corpora remain dominated by short, single-speaker utterances collected through crowdsourcing, and the largest publicly available corpus of this kind is the 229-hour Google Bangla speech corpus. The Bengali-Loop benchmark (Tabib et al., 2026) was introduced specifically to address the long-form gap and consists of two sub-corpora: a long-form ASR corpus of 191 recordings totalling 158.6 hours from eleven YouTube channels, with a reproducible subtitle-extraction and human-in-the-loop verification pipeline, and a speaker diarization corpus of 24 recordings totalling 22 hours with fully manual speaker-turn labels in CSV format. The benchmark reports that an out-of-the-box pyannote.audio pipeline reaches a diarization error rate of 40.08% on the diarization test set, and that a BengaliAI fine-tuned Whisper model reaches a word error rate of 34.07% on the ASR test set. These numbers anchor the gap between unadapted performance and the level required for downstream applications, and they are consistent with the claim that further gains on low-resource languages depend primarily on labelled data rather than on modelling innovation.

The DL Sprint 4.0 Kaggle competition, run on a subset of the Bengali-Loop data, attracted multiple community submissions that fine-tuned the pyannote segmentation backbone and the tugstugi Whisper model on the competition training set. One submission (Bhuiyan et al., 2026) reports a DER of 23.92% on the diarization track and a WER of 24.41% on the ASR track, both well below the unadapted baselines; other submissions report even lower numbers, with public-leaderboard DER around 20% (Tanvir et al., 2026) and public WER around 16% (Tanvir et al., 2026). The competition reports also produced three pieces of empirical evidence that are directly relevant to the present work: first, that fine-tuning the segmentation component of pyannote.audio is more impactful than fine-tuning the embedding model (Chowdhury and Chowdhury, 2026); second, that vocal source separation combined with silence-aware chunking reduces word-boundary errors by preserving sentence-internal structure (Chowdhury and Chowdhury, 2026); and third, that targeted data augmentation for muffled-zone robustness contributes measurable gains (Tanvir et al., 2026). Each of these findings depends on having access to labelled long-form Bangla audio, and the labelled data was produced in part by manual annotation. The competition submissions themselves include a description of the data pipeline used to construct the training set, and the manual verification of the automatically extracted transcripts and the manual annotation of speaker turns is reported as a substantial component of that pipeline. The present work is complementary to these modelling efforts: by targeting the human annotation step, it attacks the constraint that the modelling efforts also identify, namely the scarcity of high-quality annotated data.

3 Research Gap

The four threads reviewed in Section 2 have evolved largely independently. Screen-based diarization correction is mature, with both commercial and research tools that instrument correction cost (Broux et al., 2018; Tahon et al., 2024; Yamaguchi, 2026); tangible interfaces for time concern sequencing and authoring of temporal media (Jordà et al., 2007; Bennett et al., 2012; Kaltenbrunner and Vetter, 2018; Mahieux et al., 2022); and human-in-the-loop annotation has been shown to be effective when the interaction is graphical (Prokopalo et al., 2022). The token+constraint framework provides a vocabulary for tangible interaction with digital information (Ullmer et al., 2005, 2003; Ishii and Ullmer, 1997), and empirical work has demonstrated measurable differences between tangible and graphical interaction on memory and load outcomes (Patten and Ishii, 2000; Chandrasekera and Yoon, 2015).

The gap addressed in this article lies at the intersection of these areas. The specific combination of (i) passive token-based correction, (ii) of the temporal output produced by a frozen recognizer, (iii) deployed as an annotation accelerator for a low-resource language, has not been explored. No existing study asks whether the choice of interaction modality governs which class of recognizer error a human annotator catches. The closest contemporary neighbour (Yamaguchi, 2026) varies graphical assistance while holding modality constant. The present study holds the recognizer and the annotation operations constant and varies the modality.

4 Objectives

The central question is whether the interaction modality used to correct automatic speaker diarization changes which class of error a human annotator catches and repairs, and at what cost in time and effort. The study tests three hypotheses.

H1. Annotators using the tangible interface correct speaker-confusion errors, in which speech is attributed to the wrong speaker, with higher accuracy and lower correction time than with the graphical interface, because per-speaker tokens externalize identity tracking.

H2. Annotators using the graphical interface correct temporal boundary errors, that is the offsets of segment start and end points, with greater precision than with the tangible interface, because the graphical interface affords finer temporal control through waveform zoom and pointer dragging.

H3. Perceived workload and stated preference differ by modality. These are treated as secondary outcomes.

The primary prediction is the interaction between modality and error type, the double dissociation, which is more robust to novelty and engagement effects than any single main effect. A confirmatory result on the interaction, with H1 and H2 pointing in opposite directions, would establish that interaction modality governs which class of recognizer error a human annotator catches, a principle that transfers beyond diarization to other temporal labeling tasks. Detecting a crossover interaction of moderate size requires more statistical power than detecting a single main effect; the sample size in Section 5.5 is therefore set with the interaction term as the primary inference target, and a sensitivity analysis is pre-registered for the smallest effect of the predicted direction that the study can reliably detect.

5 Proposed Methodology

5.1 Tangible interface

The interface uses one passive token per speaker. Each token is a printed ArUco fiducial marker fixed to a small rigid object such as a wooden craft cube or a bottle cap, which requires no specialized fabrication. The camera reads only marker identity and position. Tokens are placed on a printed paper timeline divided into horizontal lanes. Placing a speaker's token in a region assigns that segment to the speaker, and moving the token reassigns it. Sliding a token along its lane adjusts the corresponding segment boundary, which gives the tangible condition a genuine if coarser boundary-editing capability so that the boundary comparison is not decided by capability alone.

Sensing uses a consumer USB webcam of at least 1080p resolution, mounted overhead on a copy stand and feeding directly into OpenCV. A local fixed-framerate feed avoids the latency and connection fragility of a phone-camera arrangement. Autofocus is disabled and exposure and white balance are fixed through the camera controls to prevent tracking loss. Because webcam resolution determines marker positional resolution, and therefore the smallest boundary adjustment the tangible condition can express, the camera is fixed and characterized before data collection. Marker detection is restricted to identity and position, with no rotation or multi-token grammar, which keeps the vision pipeline small and inexpensive to run.

During the tangible condition the screen presents the current waveform, segment boundaries, speaker coloring, and playback position, but accepts no input. The mouse and keyboard are removed from the workspace so that the screen cannot serve as an alternative interface, and all correction is performed through the tokens. Feedback is therefore visual rather than haptic, consistent with established tabletop tangible interfaces such as the reacTable (Jordà et al., 2007).

5.2 Graphical baseline

The graphical baseline is built by wrapping the open-source waveform library wavesurfer.js, together with its regions plugin, in a minimal interface. This reuses the most demanding parts of the frontend, including waveform rendering, zooming, playback synchronization, and draggable region handles, while keeping the comparison under experimental control. A full annotation platform is deliberately not adopted, because its accumulated features would introduce differences unrelated to interaction modality and reducing it to parity would often exceed the effort of a focused implementation.

The baseline is governed by three constraints. It exposes exactly the operations available in the tangible condition, namely reassigning a segment to a speaker by selecting its region and adjusting a boundary by dragging a region edge, and no others. It omits production features that have no tangible counterpart, including keyboard shortcuts, multi-track layouts, snapping, and autosave. It remains a competent editor, since a weakened baseline would invalidate any observed advantage for the tangible interface.

5.3 Diarization backend

A pre-trained Bangla diarization pipeline, comprising pyannote segmentation, speaker embedding, and clustering (Bredin et al., 2020; Plaquet and Bredin, 2023), produces diarization output before any session. No model runs during interaction, so there is no computational load at interaction time and laptop-grade hardware is sufficient. The model is not modified and is not part of the contribution.

5.4 Experimental control across conditions

Both conditions are implemented as a single application with a shared state model and a shared renderer, differing only in the source of input. The on-screen representation is produced by the same rendering code in both conditions, which removes visual presentation as a source of difference. In the graphical condition the input source is mouse interaction with the rendered regions. In the tangible condition the input source is the marker tracker, which emits the same logical operations of speaker reassignment and boundary movement while the rendered regions remain non-interactive. Sharing the waveform view across conditions is also what makes the boundary comparison interpretable. Because both conditions act against the same visual reference, the boundary measurement reflects the precision of the manipulation channel, token sliding against pointer dragging, rather than differences in visual access to the signal.

Playback is held constant across conditions through a single physical control used for play, pause, and seeking in both, so that playback does not introduce an asymmetry unrelated to the interaction modality.

Two consequences of strict information parity are noted. First, because speaker identity is shown on screen in both conditions, a confirmed H1 is attributable to the manipulation of identity, that is grasping and placing the correct token without visual search of a label, rather than to a stronger form of external representation. This narrows the mechanism while strengthening internal validity. Second, with a vertical monitor the locus of action on the table is spatially separated from the locus of feedback on the screen, whereas the graphical condition co-locates them. This separation is a property of the tangible setup as instantiated and is reflected in the scope of the claims, which concern token interaction on a table with screen-based feedback.

5.5 Design and participants

The study uses a within-subjects design with two conditions, the tangible interface and the graphical baseline, and counterbalanced order to control for learning effects. Each participant corrects a set of diarized audio clips under each condition. Detecting an interaction requires more statistical power than detecting a single main effect, so the study targets between 16 and 24 participants. The faculty collaborator conducts recruitment and runs the sessions.

5.6 Stimulus construction

The error distribution of the frozen model is first characterized on held-out Bangla audio, measuring the relative frequency of speaker-confusion errors and boundary errors. Clips are then constructed by injecting a controlled and balanced set of both error types onto ground-truth diarization, at rates consistent with the model's actual behavior, so that the exact errors are known in advance and matched across conditions and participants. Total error count and clip difficulty are held constant across trials, and clip sets are counterbalanced across conditions. The number of speakers is held constant to reduce variance.

5.7 Measures

The primary measures are correction accuracy for each error type, defined as the proportion of injected errors of that type that are correctly repaired, and correction time for each error type. Boundary precision is measured as the absolute timestamp error remaining after correction. Secondary measures are perceived workload, assessed with the NASA Task Load Index (Chandrasekera and Yoon, 2015) as a precedent, and stated preference.

5.8 Analysis

The primary analysis is a two-factor repeated-measures model with modality and error type as factors, applied to accuracy and to correction time. The interaction term tests the predicted dissociation. Boundary precision is analyzed separately. Secondary measures are reported with appropriate non-parametric tests.

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